

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):	
$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)
$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)
$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)	
Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.	
ê by polarization plug into templates above	

Pol.	ê	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{ } = (1 + \Gamma_{ }) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{ } ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_2}{\eta_1}$	$ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$ \tau_{ } ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R+T total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{ } = 1 - R_{ }$

Notes: sin θ_t = √μ₁ε₁/μ₂ε₂ sin θ_i; nonmag. ⇒ η₂/η₁ = n₁/n₂.

Intrinsic impedance η_i plug into table above	
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)}$	$\eta_0 = 120\pi \approx 377 \Omega$
Default μ _r = 1 (air, dielectric). Use μ _r ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.	
Low-loss correction $\eta_c \sigma/(\omega\epsilon) \ll 1$	
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$	
For Γ/τ just use η ₀ /√ε _r . The j term is for η _c , θ _η . See LOSSY MEDIA — 5 CASES.	

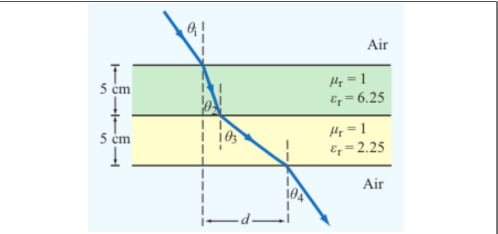
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$
Multilayer chain <i>stack of slabs 1 → 2 → 3 → 4</i>	$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical</i>	$\theta_{\text{exit}} = \theta_{\text{incident}}$
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m		
	$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$	
Type	σ/ωε	η _c , α, β
Lossless (σ=0)	0	$\eta = \eta_0 / \sqrt{\epsilon_r}$ exact $\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$
Low-loss	< 10 ⁻²	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \eta_0}{2\omega \epsilon_r}\right)$ $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: $(\eta / \sqrt{X}) e^{j\theta_\eta}$ see below
Good cond.	> 10 ²	$\eta_c \approx (1+j) \sqrt{\pi f \mu / \sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r / \epsilon_r} \eta_0$

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params $\sigma/(\omega\epsilon) \ll 1, \mu_r = 1$	
$\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), $\beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$ (rad/m), $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)	
Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$	
$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}$	
$\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$	

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$	TM mode $H_z = 0, E_z \neq 0$
Dimensions a × b with a > b along $\hat{x}, \hat{y}$; wave in $\hat{z}$.	
<i>E_x form (TE)</i> read m, n from arguments	
$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$	
<i>coef. on x = mπ/a; coef. on y = nπ/b.</i>	

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m=0 or n=0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω.

TE mode <i>generator: $\tilde{H}_z$</i>	$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \tilde{H}_y = \tilde{E}_x/Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	
TM mode <i>generator: $\tilde{E}_z$</i>	$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$
$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \tilde{H}_y = \tilde{E}_x/Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	
TEM plane wave <i>for reference</i>	$\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x / \eta$
$\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A, } k = \omega \sqrt{\mu\epsilon}$	
Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$	
$\beta = k \sqrt{1 - (f_c/f)^2} \text{ (rad/m)}$	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)}$	

CUTOFF FREQUENCY

Common cutoffs (a > b) <i>f_c formula in Table 8-3</i>	
Mode	<i>f_{mn}</i>
u_{p0}	$c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$)
TE ₁₀	$f_{10} = u_{p0}/(2a)$ (dominant)
TE ₀₁	$f_{01} = u_{p0}/(2b)$
TE ₂₀	$f_{20} = u_{p0}/a = 2f_{10}$
TE ₁₁ , TM ₁₁	$f_{11} = \frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$
Propagation requires $f > f_{mn}$. In Z _{TE} /Z _{TM} and u _g , $f_c = f_{mn}$ of the excited mode.	

ε_r FROM DISPERSION

Given ω, β, mode (m, n) <i>result is unitless</i>
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)
Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10 ⁸ (m/s), m, n (unitless).

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}$ (m/s), $u_p u_g = u_{p0}^2$
$t = L/u_g$ (s) travel time through guide of length L
Higher modes have lower u _g ⇒ arrive later.

BOUNDARY

η _i , θ _t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = Γ ²
k _i = ω√ε _{r,i} /c (rad/m)
α ₂ (Np/m), β ₂ (rad/m): lossy
z = 0 — boundary plane (m)
η ₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ _i , θ _t — angles from normal
n _i = √ε _{r,i} — index of refr.
t _i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: E ⊥ this plane
: E this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f _{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z _{TE} /TM — wave imp. (Ω)
E _x — E-field comp. (V/m)
H _y — H-field comp. (A/m)
η = η ₀ /√ε _r

VELOCITIES

u _{p0} = c/√ε _r — bulk phase vel.
u _p = u _{p0} /√1 - (f _c /f) ²
u _g = u _{p0} √1 - (f _c /f) ²
u _p u _g = u _{p0} ²
t = L/u _g — travel time
c = 3 × 10 ⁸ m/s

POWER & UNITS

%P _r , %P _t — pwr refl./trans. (unitless)
%P _r = 100 Γ ²
%P _t = 100 τ ² η ₁ /η ₂
%P _r + %P _t = 100
σ (S/m), ε (F/m), μ (H/m)
ε ₀ = 8.85 × 10 ⁻¹² F/m
μ ₀ = 4π × 10 ⁻⁷ H/m

DIPOLE TYPE BY LENGTH

STEP 0 <i>l</i> = antenna rod length (m); <i>c</i> = 3 × 10 ⁸ m/s		
λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
<i>l</i> /λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN <i>S</i> (<i>R</i> , θ), <i>D</i> = 1.5
= 1/4	Quarter-wave	ARB formula, π <i>l</i> /λ = π/4
= 1/2	Half-wave	half-wave shortcut, <i>D</i> = 1.64, <i>R</i> _{rad} = 73 Ω
= 1	Full-wave	ARB, <i>D</i> = 2.41, <i>R</i> _{rad} = 199 Ω
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — *S*(*R*, θ)

Far-field E,H phasors <i>small l</i> ≪ λ	
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$	$\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density <i>l</i> /λ < 1/50; <i>far field</i>	
$S(R, \theta) = S_{\max} \sin^2 \theta$ (W/m ²)	
$S_{\max} = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}$ (W/m ²)	
<i>l</i> — rod length (m)	<i>I</i> ₀ — peak current (A)
<i>R</i> — obs. distance (m)	θ — angle from dipole axis (rad)
<i>k</i> = 2π/λ (rad/m)	η ₀ = 120π ≈ 377 Ω
<i>S</i> _{max} — max pwr density (W/m ²)	
Max @ broadside ⊥ to dipole axis = equator plane θ _{max} = π/2; <i>D</i> exact <i>D</i> = 1.5 (1.76 dB)	
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l / \lambda)^2$ (W)	

ARBITRARY-LENGTH DIPOLE — *S*(θ)

Power density at <i>R</i> any <i>l</i>	
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$	
<i>At</i> θ = 0, π: <i>bracket</i> is 0/0. <i>L'Hôpital</i> ⇒ <i>S</i> = 0 (<i>no axial radiation</i>). <i>TI-36X Pro alt:</i> <i>evaluate bracket at</i> θ = 0.001 <i>rad</i> → <i>tiny</i> → 0.	
Normalized pattern <i>dimensionless</i> <i>F</i> (θ) = <i>S</i> (θ)/ <i>S</i> _{max}	

COMMON ANGLES (deg ↔ rad, trig values)

Conversion multiply by π/180 or 180/π						
$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$			$\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$			
°	rad	sin θ	cos θ	sin ² θ	cos ² θ	θ
0	0	0	1	0	1	
30	π/6	1/2	√3/2	1/4	3/4	
45	π/4	√2/2	√2/2	1/2	1/2	
60	π/3	√3/2	1/2	3/4	1/4	
90	π/2	1	0	1	0	
180	π	0	−1	0	1	

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

<i>N</i> identical elements along axis, spacing <i>d</i> , equal phase (broadside, θ _{max} = π/2). Total pattern <i>S</i> = <i>S</i> _e · <i>F</i> _a (<i>S</i> _e = single elem).	
Array factor (uniform, broadside) <i>k</i> = 2π/λ, γ = <i>k d</i> cos θ	
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\max} = N^2$ at θ = π/2	
Normalize $F_a^{\text{norm}} = F_a / N^2$	
HPBW (broadside, uniform) use <i>N d</i> , not (<i>N</i> −1) <i>d</i>	
$\beta \approx 0.88 \frac{\lambda}{N d}$ (rad) = 50.8° · $\frac{\lambda}{N d}$	
<i>Grating lobes appear if</i> <i>d</i> > λ. <i>Avoid</i> .	

BEAM PARAMETERS — β, Ω_p, *D*

Normalize <i>always start here</i>		
$F(\theta) = S(\theta) / S_{\max}$ (unitless)		
Beamwidth β at threshold <i>X</i>		
Set <i>F</i> (θ _X) = <i>X</i> , solve for θ _X ; β = 2θ _X (symmetric)		
Threshold	<i>X</i> = 10 dB/10	θ _X (Gauss. short-cut)
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any <i>F</i> (θ _X)	any <i>X</i>	$F(\theta_X) = X$
Pattern solid angle Ω _p <i>recognize pattern, look up</i>		
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr)		
<i>No</i> φ dep.: Ω _p = 2π ∫ ₀ ^π <i>F</i> (θ) sin θ <i>d</i> θ.		
<i>θ</i> is the integration variable — <i>DON'T</i> plug in a number for θ (e.g. θ _{1/2}). Integrate over θ ∈ [0, π].		
Pattern <i>F</i> (θ)	Ω _p (sr)	
Isotropic (<i>F</i> = 1)	4π ≈ 12.57	
Hertzian sin ² θ	8π/3 ≈ 8.38	
Half-wave dipole	≈ 7.66	
Narrow Gauss <i>e</i> ^{−<i>a</i>θ²}	π/ <i>a</i> = πθ _{1/2} ² /ln 2	
2-axis HPBW (aperture)	β _{xz} · β _{yz}	
Other (use integral)	numerical	

TI-36X Pro: [2nd] [d/dx] for $\int_{\square}^{\square}$, RAD mode, multiply by 2π.

Directivity <i>D</i> <i>always</i> <i>D</i> = 4π/Ω _p ; <i>common shortcuts below</i>	
$D = \frac{4\pi}{\Omega_p}$ (unitless)	<i>D</i> _{dB} = 10 log ₁₀ <i>D</i>
WORD TRAP: “direction” = θ _{max} (rad); “directivity” = <i>D</i> (unitless).	
<i>Dipoles</i> → COMMON DIPOLE PARAMS table. Aperture: <i>D</i> ≈ 4π <i>A</i> _p /λ ² . 2-axis HPBW: <i>D</i> ≈ 4π/(β _{xz} β _{yz}). Circular: <i>D</i> ≈ 4π/β ² .	
Gain (if asked) ξ = rad. eff., 0 < ξ ≤ 1	
<i>G</i> = ξ <i>D</i>	<i>G</i> _{dB} = 10 log ₁₀ <i>G</i>
<i>Lossless</i> / <i>ideal antenna</i> ⇒ <i>G</i> = <i>D</i> .	

COMMON DIPOLE PARAMETERS

<i>Lossless</i> (ξ = 1): <i>G</i> = <i>D</i> .				
Type	<i>l</i>	<i>D</i>	<i>D</i> _{dB}	<i>R</i> _{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	< λ/50	1.5	1.76	80π ² (<i>l</i> /λ) ²
Half-wave	λ/2	1.64	2.15	73.2
Monopole	λ/4 (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: *P*_{rad} = 36.6 *I*₀² *W*. Monopole (λ/4 above ground): same patt. as half-wave but *P*_{rad}, *R*_{rad} halved.

ANTENNA AREAS (*A*_e, *A*_p)

Effective area <i>antenna's RX cross section</i> ; <i>D</i> from BEAM PARAMS or table above	
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section <i>wire dipole, diameter d</i> ; <i>l</i> see table above	
$A_p = l d$ (m ² , any dipole)	
Inverse: <i>D</i> from <i>A</i> _e	
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
<i>Thin wire:</i> <i>A</i> _e ≫ <i>A</i> _p (e.g., Hertzian <i>A</i> _e = 3λ ² /8π ≈ 0.119λ ²).	

FRIIS TRANSMISSION FORMULA

For each antenna's G: if given by NAME (“half-wave dipole” etc.) → COMMON DIPOLE PARAMS (*G* = *D*, *lossless*). Else (dB/linear value given) → use as given (convert dB → linear).
Pwr density at RX, then RX pwr *G*_t, *G*_r linear gains; λ = *c*/*f*

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: *P*_r = *S*_r *A*_{e,r}.
Solve for *R* *max range*

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using *A*_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: *L*_{fs} = (4π*R*/λ)², so *P*_r = *G*_t *G*_r *P*_t / *L*_{fs}.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

*F*_t, *F*_r normalized patterns at the link directions; *F* = 1 when aligned.

Recipe 1. λ = *c*/*f*. 2. Convert any dB gain to linear: *G* = 10^{*G*_{dB}/10}.

3. Get *G*_t, *G*_r by antenna type:

- named dipole → COMMON DIPOLE PARAMS (*G* = *D* *lossless*)
- aperture / dish → APERTURE ANTENNAS (*G* = *D* ≈ 4π*A*_p/λ²)
- arbitrary pattern *F*(θ) → BEAM PARAMS (*G* = ξ · 4π/Ω_p)
- given in dB → dB↔LINEAR table

4. Plug into Friis.
Use linear *G*, not dB. *Convert FIRST*.

dB ↔ LINEAR

Works for any power-like quantity (*G*, *D*, *P*_r/*P*_t, *SNR*, ...):

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

dB	linear	dB	linear
----	--------	----	--------

Memorize: 3 dB ↔ ×2, 10 dB ↔ ×10. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (*power* ∝ *E*²).

NOISE & SNR

Noise power *thermal noise into matched receiver*

$$P_N = k_B T_{\text{sys}} B \text{ (W)}$$

*k*_B = 1.38 × 10^{−23} J/K; *T*_{sys} system noise temp (K); *B* receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs SNR > 10 dB for usable signal.

APERTURE ANTENNAS (horn, dish)

*A*_p physical aperture area (m²); *A*_e effective area (m², = λ² *D*/4π); β_{xz}, β_{yz} HPBWs in two planes (rad); ℓ, *d* aperture side / diameter (m).

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi / \beta^2$$

HPBW and *D* by aperture shape *uniform illum.; β in rad*

Shape	<i>A</i> _p	HPBW (rad)	<i>D</i> (unitless)
-------	-----------------------	------------	---------------------

$$\text{Rect. } \ell_x \times \ell_y \quad \beta_x \approx 0.88\lambda / \ell_x \quad D \approx 4\pi / (\beta_x \beta_y)$$

$$\ell_y \quad \beta_y \approx 0.88\lambda / \ell_y \quad = 4\pi \ell_x \ell_y / \lambda^2$$

$$\text{Square } \ell \quad \ell^2 \quad \beta \approx 0.88\lambda / \ell \quad D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$$

$$\text{Circular (diam. } d) \quad \pi d^2 / 4 \quad \beta \approx 1.02\lambda / d \quad D \approx \frac{4\pi}{\beta^2} = \frac{4\pi d^2}{\lambda^2}$$

Scaling (any ratio) *D* ∝ *A*_p/λ²; β ∝ λ/√*A*_p

$$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. *Examples:* double area → *D* × 2, β × 1/√2; double freq → *D* × 4, β × 1/2.

DIPOLE

l — antenna length (m)
 $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
 I_0 — peak current (A)
 $k = 2\pi/\lambda$ (rad/m)
 R — obs. dist. (m); θ from axis
 $\eta_0 = 120\pi \approx 377 \Omega$
 $S(R, \theta)$ — pwr density (W/m²)

BEAM

$F(\theta) = S/S_{\max}$ (unitless)
 a — coef. on θ^2 in Gaussian
 β — HPBW (rad or °)
 Ω_p — pattern solid angle (sr)
 $D = 4\pi/\Omega_p$ — directivity
 ξ rad. eff.; $G = \xi D$
 $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
 P_r — power received (W)
 G_t — transmit gain (linear)
 G_r — receive gain (linear)
 $S_r = G_t P_t / (4\pi R^2)$ (W/m²)
 $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
 $G = 10^{G_{\text{dB}}/10}$
 $3 \text{ dB} = \times 2, 10 \text{ dB} = \times 10$
 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m²)
 $A_e = \lambda^2 D / (4\pi)$ (m²)
 $A_e \approx A_p$ for aperture
 $D \approx 4\pi A_p / \lambda^2$ (unitless)
 $\beta \approx 0.88\lambda/\ell$ (rect/sq.)
 $\beta \approx 1.02\lambda/d$ (circular)
 $2A_p \Rightarrow 2D, \beta/\sqrt{2}$

ARRAY

N — number of elements
 d — spacing (m)
 $\gamma = kd \cos \theta$
 $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
 $F_a^{\max} = N^2$ at $\theta = \pi/2$
 $F_a^{\text{norm}} = F_a / N^2$
 $\beta \approx 0.88\lambda/(Nd)$ (broadside)